```
  }
}
```

❶ When the user is not authenticated, no audit metadata is saved.

❷ When the user is authenticated, the information about who created or updated the entity is correctly included in the data.

Open a Terminal window, navigate to the Catalog Service root folder, and run the newly added tests as follows:

```
$ ./gradlew test --tests OrderRepositoryR2dbcTests
```

If you experience any failures, it might be because you haven't updated the test cases where the `Order()` constructor is used. We have added new fields to the domain model, so remember to update those test cases as well.

That concludes our discussion of authentication, authorization, and auditing for both imperative and reactive cloud native applications using Spring Boot, Spring Security, Spring Data, and Keycloak.

Polar Labs

Feel free to apply what you have learned in the previous chapters and update Catalog Service and Order Service for deployment.

1. Update the Docker Compose definitions for both applications to configure the Keycloak URL. You can use the container name (`polar-keycloak:8080`), which gets resolved by the built-in Docker DNS.
2. Update the Kubernetes manifests for both applications to configure the Keycloak URL. You can use the Keycloak Service name (`polar-keycloak`) as the URL, since all interactions happen within the cluster.

You can refer to the Chapter12/12-end folder in the code repository accompanying the book to check the final results (https://github.com/ThomasVitale/cloud-native -spring-in-action). You can deploy the backing services from the manifests available in the Chapter12/12-end/polar-